

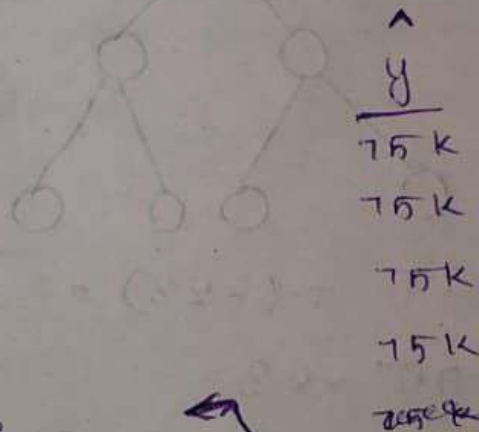
GRADIENT BOOSTING ALGORITHM

- ① Regression
- ② Classification

Regression

Dataset

<u>Exp</u>	<u>Degree</u>	<u>Salary</u>
→ 2	B.E	50K
3	Masters	70K
5	Masters	80K
6	PHD	100K
		Avg 75K



$(y - \hat{y}) \rightarrow R_1$	R_2
-25K	-23
-5K	-3
5K	3
25K	20

$\hat{y}$	R_3
72.7	-22.7
74.7	-4.7
80	-
100	-

Steps

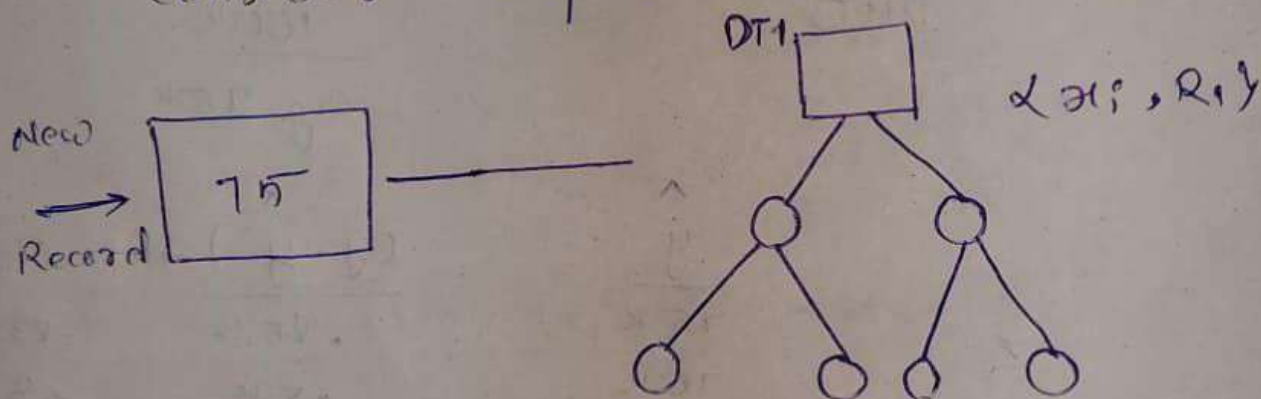
① Create a Base Model

75

$$\text{Average} = \frac{50 + 70 + 80 + 100}{4} = 75$$

② Compute Residuals, Errors

③ Construct a Decision Tree
consider input x_i and o/p R_i



$$\text{Predicted o/p} \Rightarrow 75 + (-23) =$$

$$= 75 - 23$$

$$= \underline{\underline{52}} \quad \text{Overfitting?}$$

Predicted O/P $\Rightarrow 75 + \alpha (DT_1)$

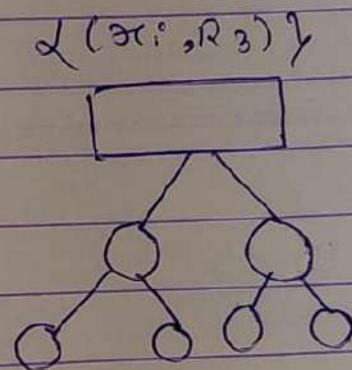
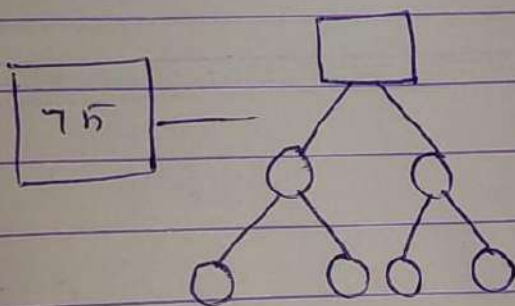
$$\alpha = \text{Learning Rate} = 75 + (0.1) (-23)$$

$$\begin{aligned} \alpha = 0.1 &= 75 - 2.3 \\ &= \underline{\underline{72.7}} \end{aligned}$$

$$\Rightarrow 75 + 0.1 (-3)$$

$$\Rightarrow 75 - 0.3$$

$$= 74.7$$



$$F(x) = \alpha_0 h_0(x) + \alpha_1 (h_1(x)) + \alpha_2 (h_2(x)) + \alpha_3 (h_3(x)) + \dots + \alpha_n (h_n(x))$$

Learning Rate

$$\alpha = 0.1$$

$$F(x) = \sum_{i=0}^n \alpha_i h_i(x)$$

$\rightarrow$ Final Function of Gradient Boosting